Customer & Sales Analytics: Findings and Method

Project SG/PRJ/DA-02 | Vaishnavi B. Gawande | Sample Superstore data, 2014-2017

1. Summary

Sales grew 51% in four years, but discounts above 20% lose money, and a fifth of revenue sits with customers who are drifting away. This report explains how I reached three findings and three recommendations using a star-schema database, SQL, RFM segments, cohort analysis, a forecast and a Power BI report.

Key findings

- Sales are growing and strongly seasonal: $484K in 2014 to $733K in 2017 (up 20% in 2017). September, November and December are the strongest months every year, and Q4 brings in about 38% of sales.
- Customers are loyal, but 22% of sales sit with customers who have gone quiet. 98.5% of customers reorder, about every 188 days. Champions and Loyal customers (329 people) bring 53% of sales; the 146 At Risk customers hold $495K and have not ordered for about 7 months. Only 11 customers were new in 2017.
- Deep discounts destroy profit: order lines discounted above 20% lost $135K while undiscounted lines earned $321K. Central discounts the most (24% average) and has the lowest margin (7.9%, against 14.9% in West).

Recommendations

- Cap discounts at 20% and require approval above it, starting with Tables and the Central region.
- Win back the 146 At Risk customers now, and follow up new customers within 60 days.
- Plan stock and staff around September to December, and plan with a forecast range: Q1 2018 sales about $148K, likely between $89K and $207K.

2. Data and model

The data is the Sample Superstore dataset: 9,994 order lines, 5,009 orders and 793 customers. I reshaped the flat file into a star schema, with one fact table and four dimensions. I chose this dataset because customers buy repeatedly (781 of 793 reorder), which RFM and cohort analysis need.

Table	One row per	Rows	Key
fact_sales	order line	9,994	Row_ID
dim_customer	customer	793	customer_key
dim_product	product (ID and name)	1,894	product_key
dim_geography	location	632	geo_key
dim_date	day, 2014 to Jan 2018	1,492	date_key

Table 1: Star schema tables.

Three data-quality findings shaped the model:

- 32 Product IDs have more than one product name, so a product is the pair of ID and name. Joining on the ID alone would duplicate revenue.
- One postal code appears with two city, state and region combinations, so a location is the full combination.
- Ship dates run to 5 January 2018, so the date table extends to 31 January 2018.

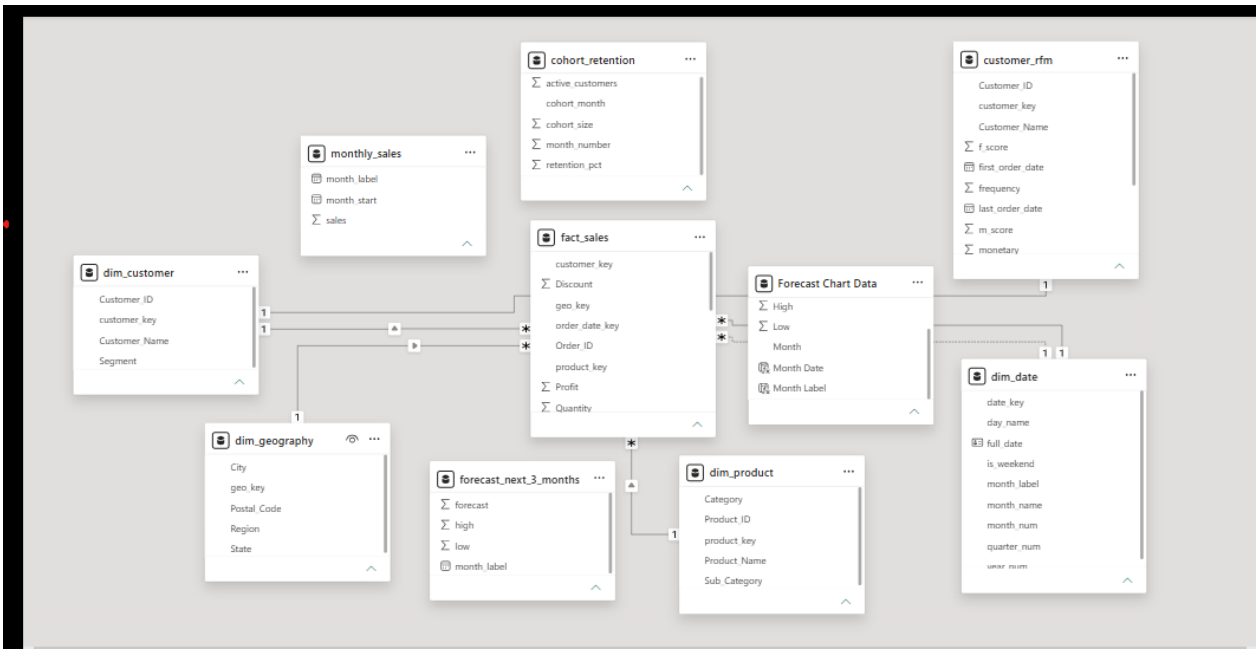


Figure 1: Data model in Power BI. fact_sales sits in the centre with four dimensions.

Row Labels	Sum of Sales
2014	4,84,247.50
1	14,236.90
2	4,519.89
3	55,691.01
4	28,295.35
5	23,648.29
6	34,595.13
7	33,946.39
8	27,909.47
9	81,777.35
10	31,453.39
11	78,628.72
12	69,545.62
2015	4,70,532.51
1	18,174.08
2	11,951.41
3	38,726.25
4	34,195.21
5	30,131.69

2017	7,33,215.26
1	43,971.37
2	20,301.13
3	58,872.35
4	36,521.54
5	44,261.11
6	52,981.73
7	45,264.42
8	63,120.89
9	87,866.65
10	77,776.92
11	1,18,447.83
12	83,829.32
Grand Total	22,97,200.86

Excel PivotTable of sales by year and month. Excel groups digits in the Indian style, so 22,97,200.86 = 2,297,200.86 and 1,18,447.83 = 118,447.83, matching SQL.

Checks

Measure	SQL	Power BI	Excel
Total sales	2,297,200.86	2,297,200.86	2,297,200.86
Total profit	286,397.02	286,397.02	-
Orders	5,009	5,009	-
Customers	793	793	-
Sales, November 2017	118,447.83	-	118,447.83
Sales, February 2014	4,519.89	-	4,519.89

Table 2: Results reconciled across SQL, Power BI and Excel. The star-schema integrity check found no fact row without a matching customer, product, location or date.

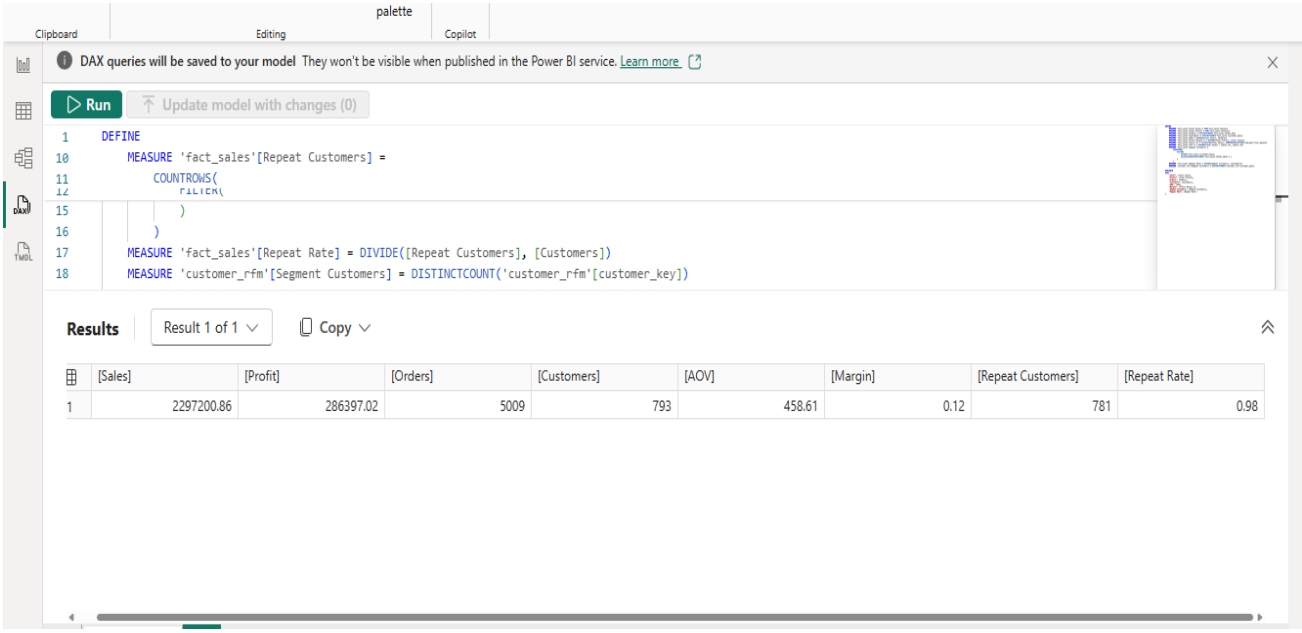


Figure 2: Power BI measures checked against SQL.

3. Method and results

3.1 SQL

Ten labelled queries use inner and left joins, GROUP BY with HAVING, CTEs, subqueries and window functions. Examples: a join with HAVING finds the three sub-categories that lose money (Tables, Bookcases and Supplies); a CTE finds the strongest months (November 2017 first, at $118,447.83); LAG gives month-over-month growth; a running total, RANK and a left join (customers with no 2017 order: 100) complete the set. Results were checked against Power BI and Excel.

3.2 Customer segments (RFM)

Each customer is scored from 1 to 5 on recency (days since last order, scored in reverse), frequency (orders) and monetary value (sales) using NTILE. Segments combine the scores.

Segment	Rule	Customers	Share of sales	Action
Champions	recency 4-5, frequency 4-5	165	29.1%	Reward, early access, referrals
Loyal	recency 3-5, frequency 3-5	164	23.5%	Upsell and bundle offers
At Risk	recency 1-2, frequency 3-5	146	21.6%	Win-back offer before they are lost
Promising	recency 3-5, frequency 1-2	140	12.9%	Encourage the next purchase
Lost	recency 1-2, frequency 1-2	167	12.5%	Low-cost reactivation
New	first order in 2017	11	0.3%	Follow up within 60 days

Table 3: RFM segments. Promising is a sixth segment added so that recent customers with few orders have a home.

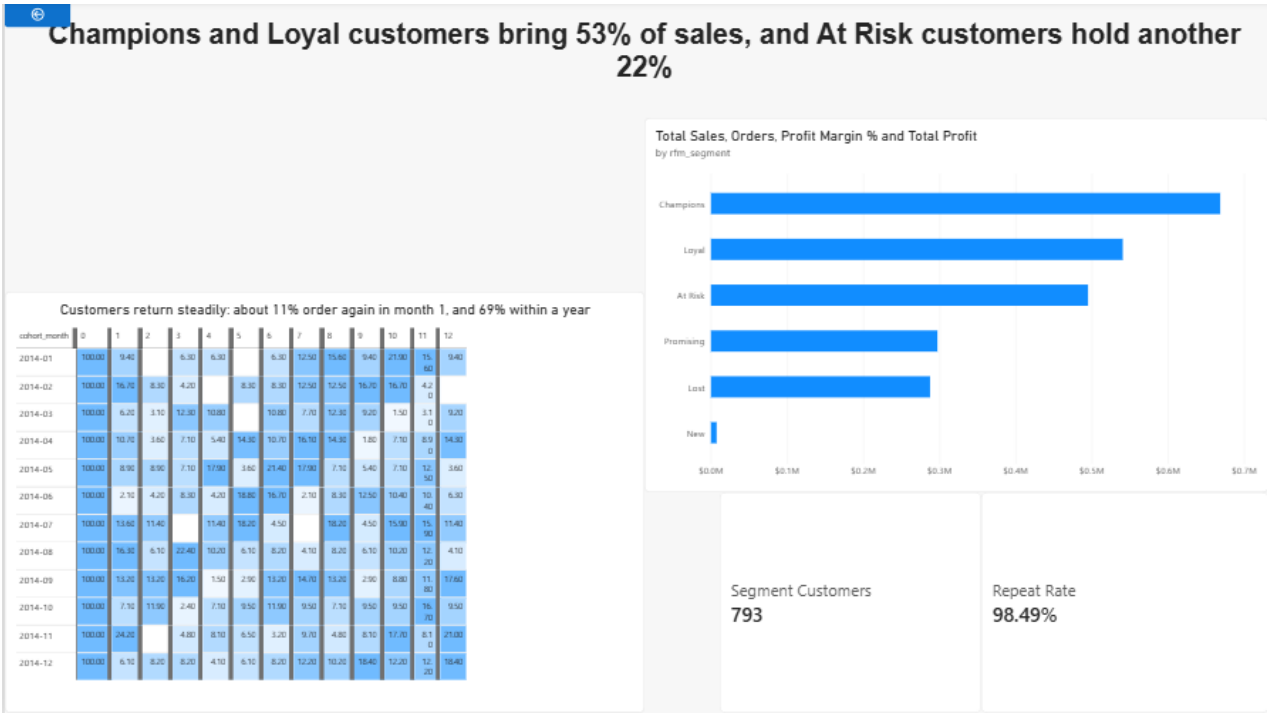


Figure 3: Segment sales and the 2014 cohort retention heatmap. Right-clicking a segment drills through to its customers (At Risk: 146).

3.3 Cohorts and retention

Customers are grouped by the month of their first order. About 11% of a cohort orders again in month 1, 24% within 3 months, 43% within 6 and 69% within 12; only 2 of the 595 customers who started in 2014 never ordered again. Customers return steadily rather than dropping off sharply, and month 2 is the weakest (6.6%), so a follow-up offer in the first 60 days is the best chance to speed up the second order. The heatmap shows the 2014 cohorts only, because 75% of customers started in 2014 and later cohorts are too small to read.

3.4 Forecast

Sales are strongly seasonal, so I tested three simple methods on the last three months of data (October to December 2017), training on the earlier months.

Method	Error (MAPE)	Rank
3-month moving average	27.5%	3
Same month last year	24.0%	2
Same month last year x growth	21.7%	1

Table 4: Forecast accuracy on the test months. MAPE is the average percentage by which the forecast missed.

Three test months are few, so I re-ran the best method on 22 earlier months. The average error was 26.3%, and about 68% of forecasts landed within 40% of the actual. I therefore show the forecast as a range of plus or minus 40%:

Month	Forecast	Range
January 2018	$52,922	$31,753 to $74,091
February 2018	$24,434	$14,660 to $34,208
March 2018	$70,856	$42,514 to $99,198
Q1 2018 total	about $148K	about $89K to $207K

Table 5: Forecast for Q1 2018 with its range.

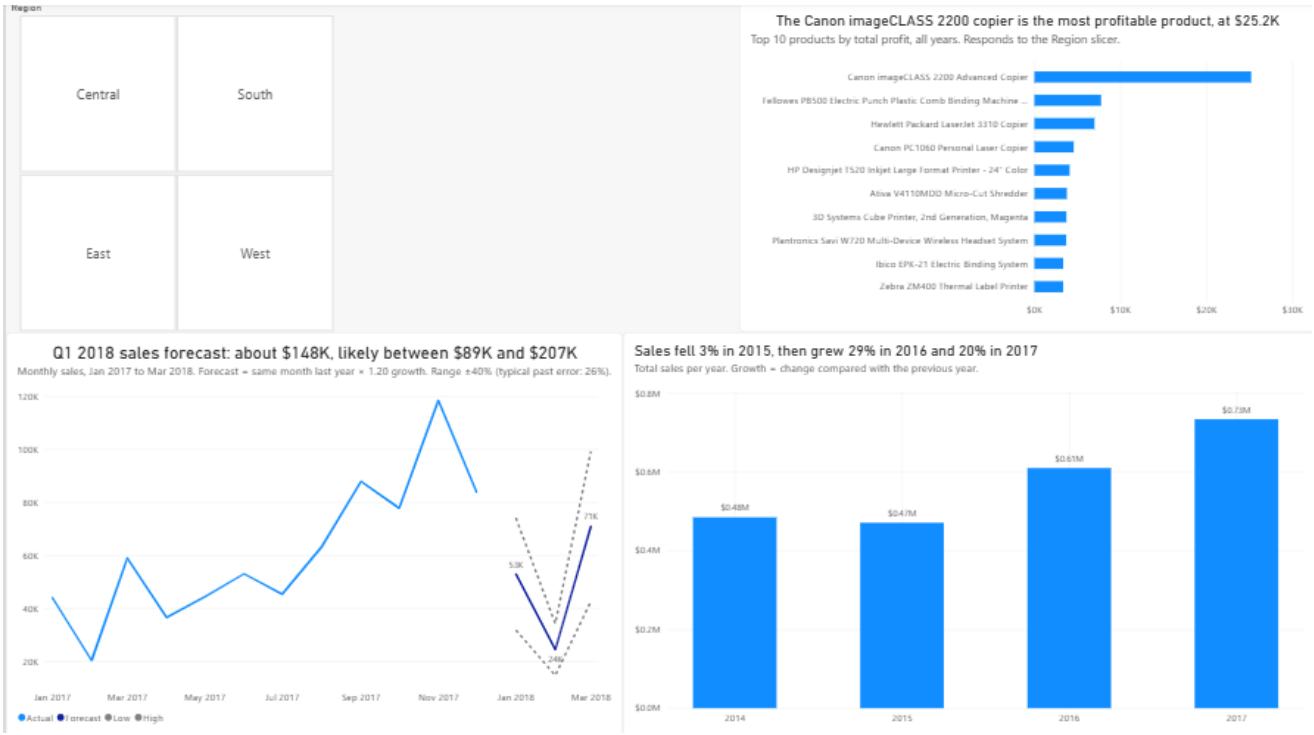


Figure 4: Forecast with range, top products by profit, and yearly growth.

3.5 Power BI report

The report is built on the star schema, not a flat table. Eleven DAX measures cover sales, profit, orders, customers, average order value, margin, year-over-year growth, repeat rate and customers by segment. Three pages (Executive summary, Customers with a drill-through to a customer list, and Products and forecast) use slicers, and chart titles state the finding.

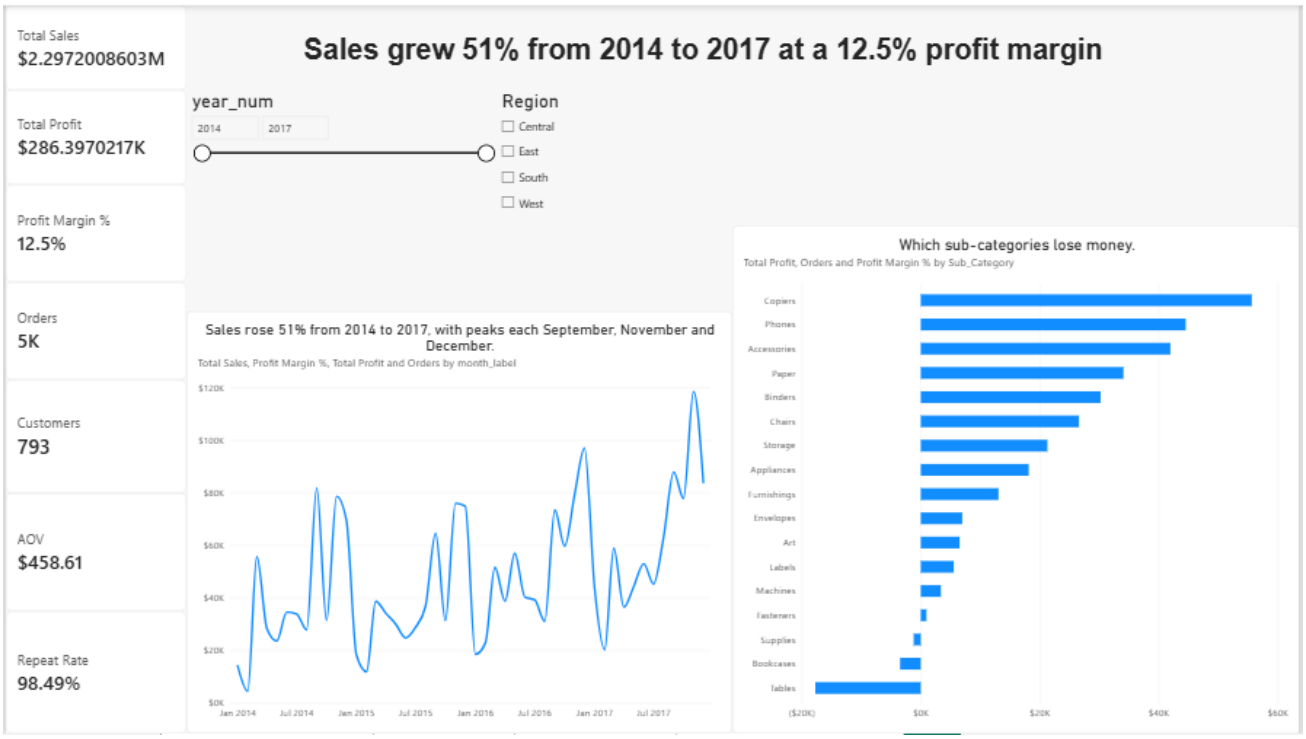


Figure 5: Executive summary page, all regions.

4. Limits

- This is sample data, not live company data, and covers four years.
- Discounts and losses occur together. This shows correlation, not proven cause.
- The forecast is wide (typical error 26%), so use the range, not a single number.
- Most customers began in 2014, so conclusions about new customers rest on few people (11 in 2017).
- The repeat rate is high (98.5%) because customers in this dataset average about 6 orders over four years; I would expect a lower rate on live data.